Zichen (Warren) Song

Singapore/Shanghai | (+86) 15398078036 | e1127317@u.nus.edu | WeChat: Szc1123581321 | github/ZCWarren

Education

National University of Singapore (NUS): Master of Science in Physics for Technology 2023/08 – 2025/01
GPA: 4.38/5, Focus on Trapped-Ion Quantum Computing; built an FPGA-based fast qubit-state detection system.

China University of Mining and Technology (CUMT): B.Sc. in Optoelectronic Info. Sci. & Eng. 2019/09 – 2023/06
GPA: 3.7/4 (Top 15%), multiple academic scholarships. Thesis: Statistical Physics – Surface Growth Dynamics.

Work Experience

Shanghai Yuliangsheng Technology, Algorithm & Modeling Engineer, Computational Lithography 2025/03 – Present

- Develop **SMO** (Source Mask Optimization) algorithms in **C++ on Linux** and drive commercialization; work across **DFM/OPC** application and fab exposure & metrology validation; fab-facing owner of SMO, supporting on-site validation at customer fabs.

Project Experience

SMO (Source Mask Optimization) Engine Development Algorithm & Feature Development / C++, Python

- Develop core SMO stages in **C++ on Linux**; routinely diagnose **process-window (PW)** optimization issues and feed fab exposure/metrology findings back to SMO/OPC R&D.
- Built the **Good Initial Illumination** stage, deriving SMO's initial illumination from diffraction patterns; developed **Job Inherit**, so new jobs can build on prior stage outputs.
- Developed the **Pattern Selection** stage (Python), one of SMO's core stages: frequency-based layout screening that matches dominant frequencies and harmonics to select representative cell windows; **shipped in the product**.

Test Pattern Generator (TPG) Sole Architect, Developer & Maintainer / Python

- Designed a **plugin-based architecture** and maintained it for a year; built NumPy-vectorized geometry algorithms and the Pattern Plugin system.
- Generates multiple layout types with matching cell windows, metrology gauges and Clip/HSS files; provides GUI, scripting and external APIs; supports **advanced-node mask making**.
- Delivered to OPC, SMO R&D and FAE teams and to limited customer use; three-level validation via self-built cases, QA verification and blind tests on real FAE data.

AI-Assisted SRAF Generation & CTM Acceleration Prototype Training & Integration / PyTorch, BicycleGAN

- Trained a **BicycleGAN** SRAF generation model and integrated it into a prototype; cut CTM runtime by **over 50%** offline (not yet in production).

Fast Qubit State Measurement and Feedback System 2023/09 – 2024/12
Centre for Quantum Technologies, NUS / Master Thesis

- Built an ion-state measurement and feedback platform with EMCCD & FPGA (DE10-Standard); used Python (Otsu thresholding, Gaussian filtering) for **multi-ion image processing**, photon statistics and thresholding; implemented state discrimination in Verilog; trained a custom **YOLOv8** model for single-ion detection.

Cosmic-Ray Propagation Simulation (TDLI, SJTU, summer): PeV density & energy spectra 2024/06 – 2024/08

Tech Skills

Languages: Proficient in Python, C++; familiar with C, Verilog; basic Rust, Swift, JavaScript, SQL.
Work Flow: Linux, Shell, Vim, Git, Git-mm, Codex, Cursor, DeepSeek Harness, Kimi Code.
Others: Hands-on experience with Transformers and GANs; familiar with HPC and distributed clusters.

Additional Information

- IELTS 7.0, CET-6 558, fluent English; strong technical writing; captain of college and company football teams with multiple championships.